

Implementation of Failure Recovery in Emergency IoT Applications Using D2D in LoRa/LoRaWAN

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Abstract—As the number of connected devices is increasing, several new communication technologies are getting developed and deployed to serve the rising need. Cellular Internet of Things (C-IoT) is the umbrella of Low Power Wide Area Network (LPWAN) technologies introduced by 3GPP in order to support critical IoT applications. C-IoT technologies are deployed by Mobile Network Operators (MNO) of the country to support IoT devices. Unfortunately, just like other wireless communication technologies, C-IoT may also suffer coverage outages in some regions of the country (e.g., Forests, basements, etc.). Unlike other wireless technologies such as LTE, 5G, etc. C-IoT is typically only deployed in one frequency spectrum (band 20 in Denmark [17]), making the handover of these IoT devices in an outage scenario difficult. This significantly affects critical IoT applications like remote health monitors, location tracers, etc., where communicating data through the network is extremely important. This paper focuses on using Device-to-Device (D2D) communication using LoRa and LoRaWAN to create a low-cost, easy-to-deploy, and manage network infrastructure to minimize data loss because of outage scenarios. In order to validate this approach, authors have developed a number of prototype devices based on C-IoT and LoRa/LoRaWAN and performed experiments.

Index Terms—Cellular IoT (C-IoT), Device-to-Device communication, LTE-M, NB-IoT, Critical IoT

I. INTRODUCTION

Internet of things (IoT) is helping in the development of the health sector, asset tracking, risk assessment, monitoring, and control system. There are multiple wireless technologies available to support IoT applications. NB-IoT and LTE-M are the Low Power Wide Area Network (LPWAN) technologies based on cellular technology known as cellular IoT (C-IoT) and operate on licensed frequency bands. Technologies like SigFox and Long Range Wide Area Network (LoRa/LoRaWAN) work on unlicensed frequency bands and are known as unlicensed LPWAN technologies. These LPWAN technologies have low power consumption, which contributes to higher battery life and has better coverage compared to WiFi. However, these LPWAN technologies have some drawbacks. LPWAN wireless technologies can not be used for higher data rate applications since they support lower data rates; along with this, unlicensed LPWAN technologies have higher latency than C-IoT.

LoRaWAN is one of the unlicensed LPWAN technology that can be deployed privately because of its simple architecture and requires a low expenditure compared to other LPWAN technologies. The coverage range of LoRa is about 5 to 15

Km in suburban areas. One LoRa gateway can serve 1000s end devices [1]. The technical specifications and standards are regulated by the LoRa Alliance [2]. LoRa is planning to acquire 75% of the IoT market by deploying satellite constellations, and the rest of the 25% market is covered by 5G cellular technology for the applications that demand higher data rates and lower latency [3]. LoRa/LoRaWAN-based business market is predicted to reach \$6.2 billion by the end of 2026, and the estimated growth is forecasted at a CAGR of 36.5% [5]. Figure 1 shows the LoRaWAN global market, where network deployment is one of the key features of this technology. Deploying a private network using LoRaWAN is less expensive and less complicated. Table I shows the cost comparison for LoRaWAN and cellular IoT.

TABLE I
COST COMPARISON OF END DEVICES FOR LORAWAN AND C-IOT [4]

Requirements	LoRa/LoRaWAN	C-IoT
Manufacturing of Chip	\$1 to \$2	\$5 to \$10
End-Device	\$4 to \$6	\$6 to \$12
License	Free	Cellular Licenses

Going through the above-mentioned ongoing development related to LoRa and LoRaWAN motivated us to investigate further to utilize this LPWAN technology as the fallback network technology in case of failure recovery for emergency IoT applications using the Device-to-Device LoRa/LoRaWAN mechanism. This work aims to leverage the LPWAN technology such as LoRa and LoRaWAN to enhance and support the connectivity for the IoT applications using the D2D system where other wireless technologies such as WiFi and cellular IoT are not available.

This paper follows the following structure: Section II gives information about previous research works and findings about LoRa/LoRaWAN D2D communication. Theoretical information about LoRa and LoRaWAN is presented in section III. Test set-up and working of this practical work are elaborated in section IV. Section V analyzes the result, and the conclusion is presented in section VI.

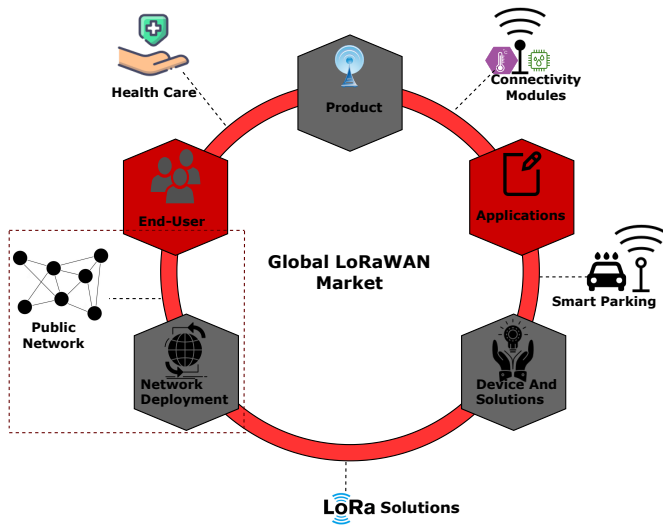


Fig. 1. Global LoRaWAN Market [6]

II. RELATED WORK

This section presents current research work and findings associated with LoRaWAN D2D applications. In the paper, [8] authors have implemented the LoRaWAN D2D communication and discussed the pros and cons of this mechanism. They have also proposed a network-assisted communication protocol for the D2D LoRaWAN. Jebril et al. [9] have developed a system using LoRa to monitor mangrove forests and transferred the captured image over LoRa physical layer using D2D communication. Using this technique, they have extended the coverage and curtailed the deployment cost of the monitoring system. A secure device-to-device link validation is proposed in the paper [10]. Authors have introduced a system to exchange SecureD2DReq and SecureD2DAns messages between the sending end-node and network server. In this system, the end nodes use the cryptographic keys to authenticate the end nodes. In paper, [11] authors have performed practical work to measure the latency for the LoRa D2D communication between 0 to 110 meters over different spreading factors (SF). This experiment concludes that when SF is 7, bandwidth is 500 kHz, and when both the devices are deployed at the same location, i.e., 0 m, the latency is 33.67 ms in that scenario. When end nodes are deployed 110 meters apart, in that case, the latency is 53 ms. The authors propose an Emergency Communication System in [12] based on D2D LoRa communication. Authors have used a system known as 'LOCATE' that includes a mobile application, and using this system, emergency data/messages can be transmitted over multi-hop LoRa links. Network-Coded Co-operation (NCC) LoRa system is elaborated in paper [13]. Using this system, end devices are able to interchange messages over D2D communication, along with this, multiple frames can transfer to a common LoRa gateway. Work conducted in this paper focuses on using Multi-RAT D2D approach to get critical information across the network to the application server.

III. LORA AND LORAWAN TECHNICAL OVERVIEW

LoRaWAN is an open source low-power network protocol that allows battery-powered IoT devices to send data to an application server. LoRaWAN uses Long Range (LoRa) wireless communication standard, a proprietary physical layer protocol developed and owned by Semtech. LoRa modulation uses the chirp spread spectrum as the preferred modulation technique. LoRa modulation, due to many of its advantages such as bi-directional communication, end-to-end stability, long-range transmission, etc., has been used in many military applications. LoRaWAN allows Mobile Network Operators (MNO) to deploy their end-to-end network infrastructure in a given geographical region. In LoRaWAN, it is also possible to deploy individual private end-to-end architecture. One of the major drawbacks of LoRaWAN is that it operates in an ISM band (800MHz) that restricts the frequency of data transmission from LoRaWAN devices [14]. Figure 2 shows the LoRaWAN architecture:

- End-device: End-devices are usually equipped with sensors, actuators, etc., collecting information from the surroundings and sending it to a server. According to the LoRaWAN specifications, it supports three different classes of devices: Class A, Class B, and Class C. All these classes have been designed to accommodate various kinds of IoT applications and meet their network requirements.
 - Class A: This device class supports uplink and downlink transmission. The device sends an uplink message and opens up two consecutive receive windows. The first window is opened after 1s on the uplink frequency, and the second window is opened after 2s on the default transmission frequency.
 - Class B: This class of devices is rather an extension of Class A. This class of devices is used when there is a frequent downlink transmission to the device. The uplink transmission happens in the same fashion as that of Class A. However, in the case of a downlink transmission, the Class B device listens for a beacon frame from the gateway to register itself for the downlink messages.
 - Class C: This class of devices is mainly for applications with no power restriction on the device. These devices are always listening on the default frequency channel. The uplink transmission of Class C devices is very similar to that of Class A and Class B devices, and it constantly listens for the downlink message.
- Gateway: The primary function of a LoRaWAN gateway is to list all the incoming messages on different LoRaWAN channels. After receiving these messages, the LoRaWAN gateway adds some additional metadata to the payload (e.g., timestamp, gateway id, spreading factor, etc.) and sends it to the network server. LoRaWAN gateways often use an IP link to communicate with the network server. Unlike in traditional mobile communication, in LoRaWAN, the end devices are not attached to a particular gateway. After the message is sent from

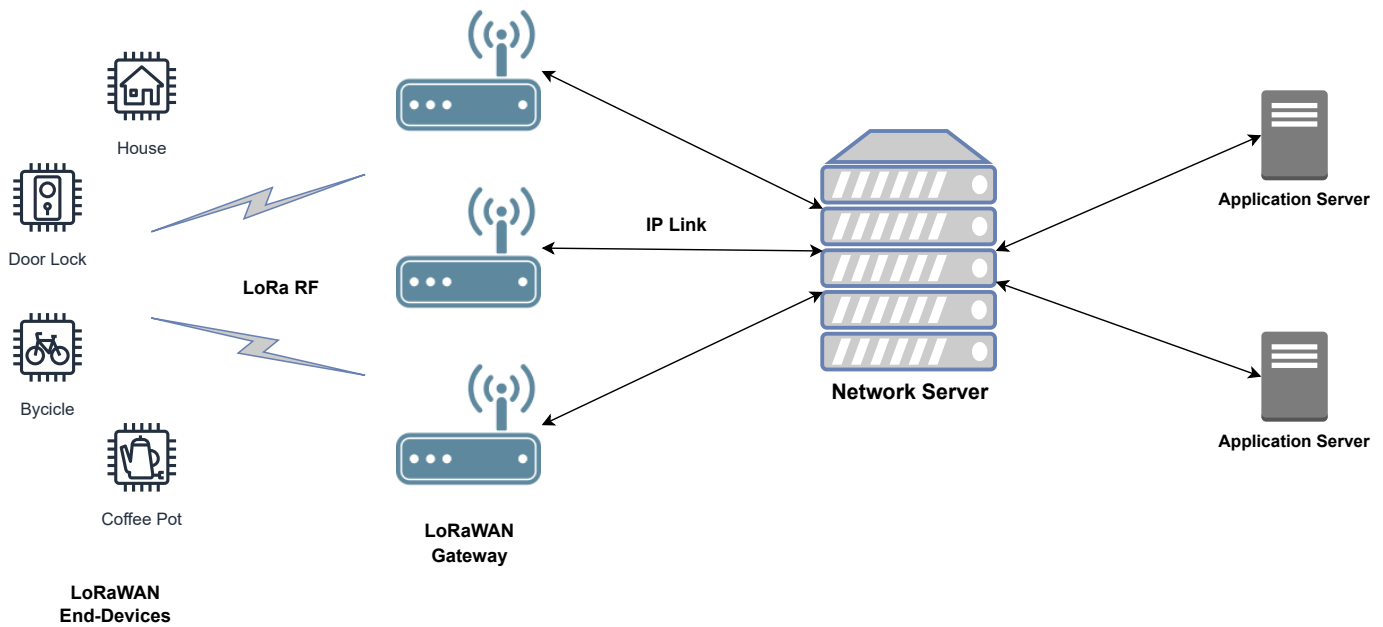


Fig. 2. LoRaWAN Architecture (based on [15])

the device, all the available neighboring gateways will receive the message, and all of them will forward the same message to the network server.

- **Network Server:** The network server in LoRaWAN functions as the core of the LoRaWAN infrastructure. It accepts all duplicate packets from the gateway, stores the metadata along with the payload, and forwards the payload to a remote application if required. LoRaWAN V1.1 also supports geolocation detection using lateration method, authenticates end devices, sends any downlink transmission to the device if available, etc. [16].
- **Application Server:** The application server is usually not part of the LoRaWAN infrastructure and is controlled by the MNO. Application developers usually deploy and maintain this server which may have algorithms to understand the information sent by the end device, visualize the data, generate any alerts, etc. The application server can send a downlink message to the end device if any action needs to be performed based on the data received. In this case, the downlink frame will be sent to the network server, which will then be queued or sent directly to the end device.

IV. TEST SETUP AND WORKING

This section describes the network architecture and the test setup developed for implementing a fallback communication procedure for C-IoT using LoRa/LoRaWAN. The designed use case focuses on the outage scenario, where there is no available network coverage from the primary communication technology, in this case, C-IoT (i.e., NB-IoT and LTE-M).

A. Network Setup Architecture

Figure 3, shows the complete end-to-end test architecture for a critical IoT application using C-IoT as the primary communication technology. The communication setup is divided into two cases, one is where the device is in a good network coverage area where all the information is transferred using the primary communication technology, and the second is when the device is in a coverage outage scenario, where the information is transferred using fallback technology. In case of good coverage, the data generated by the device is sent using LTE-M to the application server via the internet. The data travels through the MNOs network and the device constantly measure the radio conditions. When the device is powered-on for the first time, it follows the initial D2D setup procedure, described in figure 4. It is a straightforward procedure to get the available LoRa frequencies in the surroundings.

- 1) End-device shares its geolocation coordinates with the application server using REST API.
- 2) The location of the device is used by the application server to compile a list of LoRa D2D devices around the location of the device. The number of devices in the reply from the server depends on the predefined radius parameter set in the application server.
- 3) If there is a change in location, the device sends a location update to the server to which server follows the procedure in step 2 and sends a new set of LoRa frequencies back to the device.

When the device enters a coverage outage zone that cannot attach to the C-IoT network, it undergoes an outage scenario communication sequence to try and reach the application server using the fallback technology. Figure 5 describes the procedure in details.

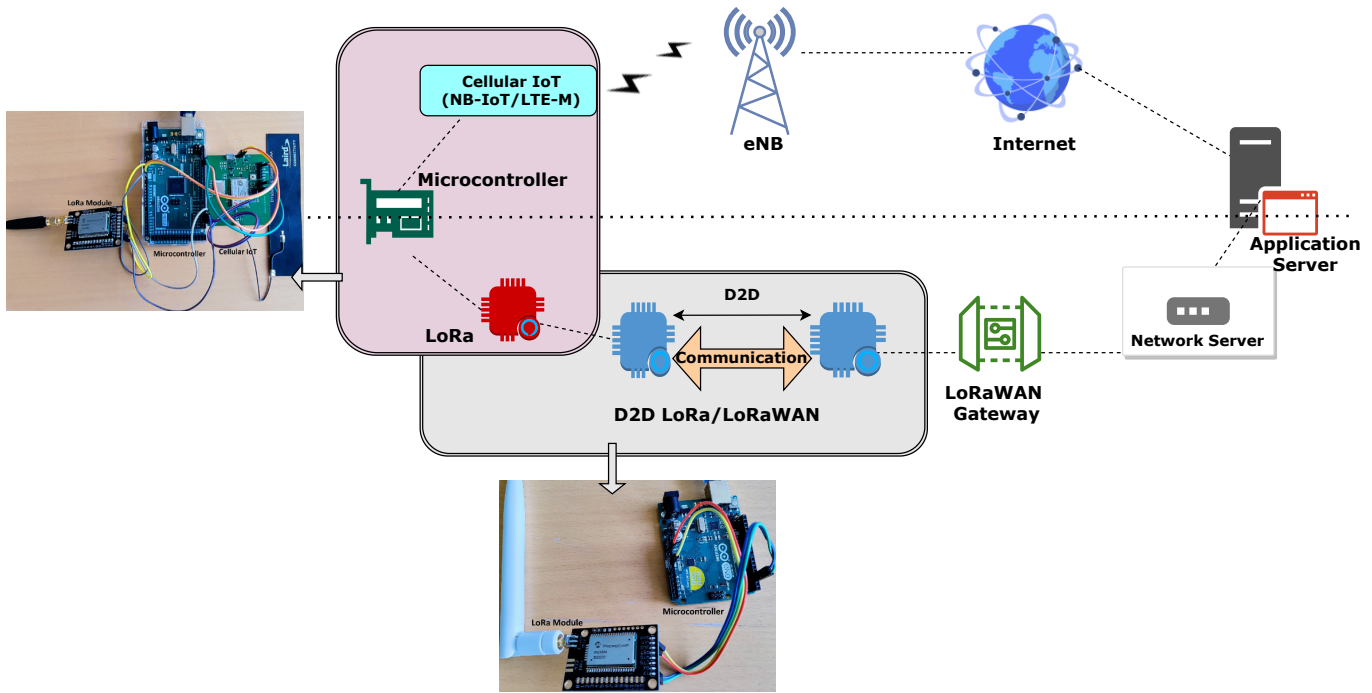


Fig. 3. Device to Device Experimental Set-Up

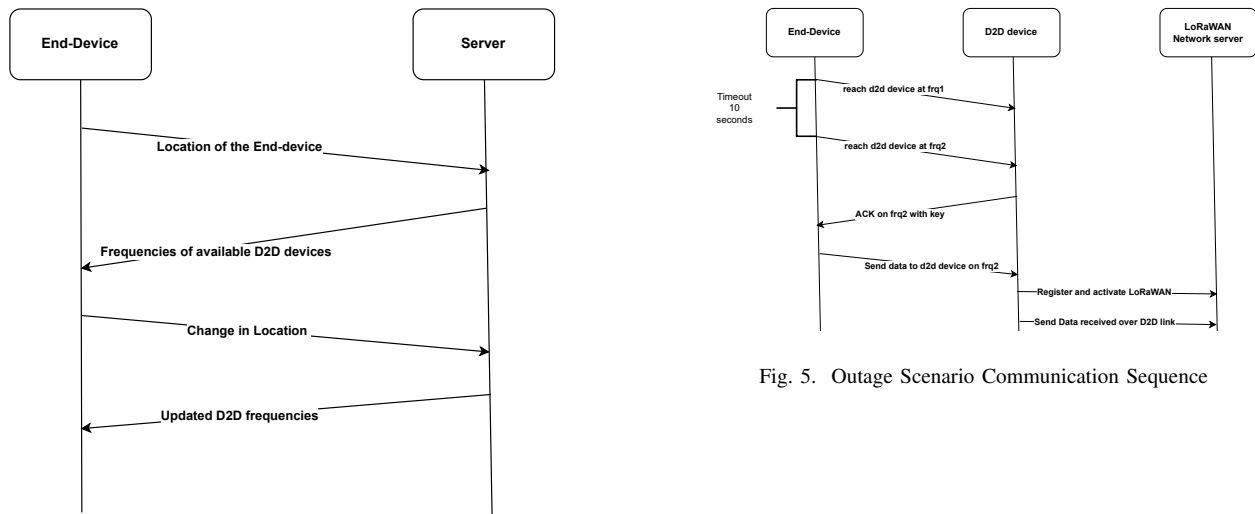


Fig. 4. Initial D2D setup

- 1) The device has an updated list of stored LoRa frequencies which have been received from the application. The device sends "hello" messages one-by-one on different frequencies from the list and waits for the LoRa device on the other side to reply. The IoT device has a pre-defined 10-second interval before switching from one frequency to another.
- 2) The LoRa D2D device, after receiving the "hello" from the IoT device, sends back an acknowledgment and a secret message. The IoT device uses the secret message to validate that the end device is a genuine endpoint to

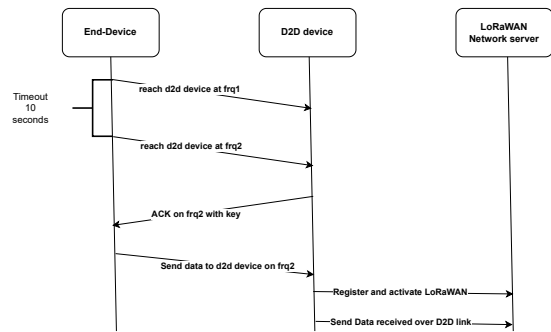


Fig. 5. Outage Scenario Communication Sequence

send critical data.

- 3) Once the message received from the LoRa D2D device is validated by the IoT device, and it will then forward the actual data over the same frequency channel.
- 4) After the data is received by the LoRa D2D, the device will change the operation mode to LoRaWAN and authenticate itself with the LoRaWAN network server. Once the authentication is complete, the LoRa D2D device will send the data from the IoT device to the network server over LoRaWAN. Depending on how the network server is configured, the received data can be forwarded to the application server and processed.
- 5) After the LoRaWAN transmission is complete, the LoRa D2D device returns to LoRa mode from LoRaWAN.

Device EUI	Local time	Freq [MHz]	Data rate	RSSI (dBm)	SNR (dB)	FCntUp	Port	IoT device data at Network Server Payload
0004A30B00FE058E	2022-06-26 18:19:55.482	867.9	SF12 BW125 4/5	-90	15	0	4	64 61 74 61 5f 64 31
0004A30B00FE058E	2022-06-26 17:52:22.776	867.5	SF12 BW125 4/5	-95	1	0	4	64 61 74 61 5f 64 31
0004A30B00FE058E	2022-06-26 17:34:54.928	867.7	SF12 BW125 4/5	-97	-22	0	4	64 61 74 61 5f 64 31

Fig. 6. Reply from the application server with a list of LoRa D2D devices

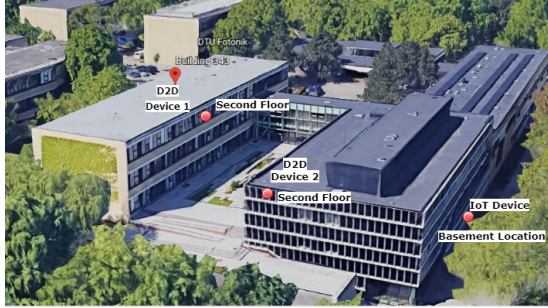


Fig. 7. Location of Deployed Devices



Fig. 8. Reply from the application server with a list of LoRa D2D devices

VI. RESULTS AND DISCUSSION

This section describes the tests conducted to validate the functionality of the end-to-end setup. The test was conducted at the Technical University of Denmark, Lyngby campus. Figure 7 shows the different locations where IoT and LoRa D2D devices were installed across the university buildings. In order to emulate the real-world use case, the IoT device was placed in the cellar of the building where the radio conditions are challenging. Two LoRa D2D devices were placed, one on the second floor of the same building, whereas the second device was placed on the second floor of the opposite building. The IoT device was programmed to follow the sequence described from figure 4 and figure 5.

Figure 8 shows the validation reply from the application server to the device with the list of frequencies. As shown in the diagram, the IoT device sends its device ID and current location to the application server as REST POST. In return, the server replied with the four frequency values from f1 to f4.

In order to validate the D2D features implemented, a hypothetical scenario has been set up in which it is assumed

that there is no coverage available at the IoT device's location. Because of this assumption, the IoT device is forced to follow the D2D outage scenario communication sequence from figure 5 and send information to the application server by switching to the fallback technology. Figure 6 shows the screen dump from the LoRaWAN network server, where it can be clearly seen that the data sent by the IoT device has been forwarded to the network server. In order to distinguish which of the LoRa D2D device is being used for the transmission of the payload, the last byte of the data was changed. In the figure, the payload received in three different tests is the same, which is 646174615f6431, which is the HEX representation for data_d1. This indicates that for all three test scenarios, the IoT device has used LoRa D2D device 1 for forwarding the critical payload to the server. This is especially impressive considering the distance between the IoT device and D2D Device 1 is more than that of the D2D Device 2. This gives a slight inside into the range of LoRa D2D link.

VI. CONCLUSION

In this paper, the authors have demonstrated an experimental setup for failure recovery in emergency IoT applications with the help of LoRa/LoRaWAN as fallback technology. The intention was to build a reliable, easy-to-scale, easy-to-maintain, parallel network infrastructure that can help transmit critical information to an application server even if the primary communication technology fails. In order to achieve this, LoRa and LoRaWAN are perfect since they offer the flexibility of using LoRa modulation to establish communication between two LoRa devices and exchange information between them without the involvement of a network. LoRa and LoRaWAN work on ISM bands, an unlicensed frequency spectrum that can be used, with some duty-cycle restrictions for any application. Another advantage of LoRa and LoRaWAN is that the cost of deployment is comparatively low. The authors have also validated the working of the end-to-end setup by performing real-world experiments. As per the preliminary results obtained during the testing and validation of the experimental setup, using a combination of LoRa physical and LoRaWAN as fallback communication technologies appear as a very promising solution. In order to deploy it as a fallback technology for critical applications, extensive testing of this experimental setup is required. The authors hope this paper can serve as a good starting point in investigating the use of LoRa and LoRaWAN as complimentary D2D technology in critical use cases.

REFERENCES

- [1] "LoRa | Advantages of LoRaWAN | Disadvantages of LoRaWAN" <https://www.rfwireless-world.com/Terminology/Advantages-and-Disadvantages-of-Lora-or-LoRaWAN.html> (Access on:29 July 2022)
- [2] "LoRa Alliance®" <https://lora-alliance.org/> (Access on:29 July 2022)
- [3] "LoRa shoots for 75% of IoT market, versus 25% for 5G; aims for stars with satellite constellation" <https://lora-alliance.org/lorawan-news/lor-shoots-for-75-of-iot-market-versus-25-for-5g-aims-for-stars-with-satellite-constellation/> (Access on:29 July 2022)
- [4] "LoRaWAN vs NB-IoT: A Comparison Between IoT Trend-Setters" <https://ubidots.com/blog/lorawan-vs-nb-iot/> (Access on:29 July 2022)
- [5] "LoRa and LoRaWAN Devices Market Overview" <https://www.industryarc.com/Report/19424/lor-a-and-lor-awan-devices-market.html> (Access on:29 July 2022)
- [6] "Global LoRaWAN Market Ecosystem; By Network Deployment, By Hardware, By Application, By End-User (Agriculture, Utilities, Consumer Electronics, Healthcare, Manufacturing, Others (Transportation, Logistics, Government)); By Region - Forecast to 2026" <https://www.alltheresearch.com/report/411/global-lorawan-market-ecosystem> (Access on:3 Aug 2022)
- [7] "Who needs cellular IoT?" Online: <https://www.firstpoint-mg.com/blog/who-needs-cellular-iot/> (Access on:26 July 2022)
- [8] K. Mikhaylov, J. Petäjäjärvi, J. Haapola and A. Pouttu, "D2D communications in LoRaWAN Low Power Wide Area Network: From idea to empirical validation," 2017 IEEE International Conference on Communications Workshops (ICC Workshops), 2017, pp. 737-742, doi: 10.1109/ICCW.2017.7962746.
- [9] Jebri, A.H.; Sali, A.; Ismail, A.; Rasid, M.F.A. Overcoming Limitations of LoRa Physical Layer in Image Transmission. *Sensors* 2018, 18, 3257. <https://doi.org/10.3390/s18103257>
- [10] J. Kim and J. Song, "A Secure Device-to-Device Link Establishment Scheme for LoRaWAN," in *IEEE Sensors Journal*, vol. 18, no. 5, pp. 2153-2160, 1 March 2018, doi: 10.1109/JSEN.2017.2789121.
- [11] N. Saqib, K. F. Haque, K. Yelamarthi, P. Yanambaka and A. Abdelgawad, "D2D-LoRa Latency Analysis: An Indoor Application Perspective," 2021 IEEE 7th World Forum on Internet of Things (WF-IoT), 2021, pp. 29-34, doi: 10.1109/WF-IoT51360.2021.9595324.
- [12] L. Sciallo, F. Fossemo, A. Trotta and M. Di Felice, "LOCATE: A LoRa-based mObile emergenCy mAnagement sysTEm," 2018 IEEE Global Communications Conference (GLOBECOM), 2018, pp. 1-7, doi: 10.1109/GLOCOM.2018.8647177.
- [13] L. H. de Oliveira Alves, J. L. Rebelatto, R. D. Souza and G. Brante, "Network-Coded Cooperative LoRa Network With D2D Communication," in *IEEE Internet of Things Journal*, vol. 9, no. 7, pp. 4997-5008, 1 April 2022, doi: 10.1109/JIOT.2021.3107784.
- [14] LoRa Alliance: "What is LoRaWAN Specification" Online: <https://lora-alliance.org/about-lorawan/> (Accessed on: 26 July 2022)
- [15] "The Things Network:LoRaWAN Architecture" Online: <https://www.thethingsnetwork.org/docs/lorawan/architecture/> (Accessed on: 26 July 2022)
- [16] LoRa Alliance: "LoRaWAN® Specification v1.1" Online: https://lora-alliance.org/resource_hub/lorawan-specification-v1-1/ (Accessed on: 26 July 2022)
- [17] GSMA: "Mobile IoT Deployment Map" Online: <https://www.gsma.com/iot/deployment-map/> (Accessed on: 26 July 2022)